Tensor Network Simulation of the Heisenberg Model on Heavy-Hex Lattices

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July 10, 2026

Abstract

We extend CppSim, a high-performance C++/HIP tensor network simulator, to the heavy-hex lattice geometry used in superconducting quantum processors. The heavy-hex graph — a honeycomb of degree-2 and degree-3 sites with a natural 3-color gate schedule — is implemented as a drop-in grid class with no changes to the gate application or belief propagation (BP) kernels. We simulate the isotropic Heisenberg model $H = J\sum_{\langle ij\rangle}(\mathbf{X}_i\mathbf{X}_j + \mathbf{Y}_i\mathbf{Y}_j + \mathbf{Z}_i\mathbf{Z}_j)$ in the Heisenberg picture using Pauli transfer matrices, and study the autocorrelation $C(t) = \langle Z_c(t)Z_c(0)\rangle$ and operator lightcone on the `heavy_hex_3x3` graph (68 sites, 76 bonds). We demonstrate chi convergence at $\chi = 200$ ($|\text{error}| < 10^{-4}$, reference $\chi = 430$) on a 32 GB GPU, and show a $\sim$10$\times$ speedup over a Julia/AMDGPU.jl reference at saturated bond dimension.

1 Introduction

Heavy-hex is the lattice topology used in superconducting quantum processors [**?**]. Each qubit site has degree 2 or 3, and the graph admits a natural 3-coloring of its bonds, enabling parallel gate application within each color class.

Classical tensor network (TN) simulation of quantum circuits on heavy-hex graphs serves two purposes: (i) validation of quantum computations for small system sizes where exact simulation is feasible, and (ii) computation of noise-free reference values for quantum error mitigation protocols such as the Multi-Product Formula (MPF) [**?**].

In this paper we show that CppSim [**?**], originally developed for rectangular 2D Ising dynamics, extends naturally to heavy-hex topology with minimal code changes. The hot-path kernels — gate application, BP, workspace management — are topology-agnostic and require no modification. Only the grid construction and Hamiltonian gate matrix need to be defined for the new geometry.

2 Heavy-Hex Lattice and Hamiltonian

We describe the graph topology, its implementation in CppSim, and the Hamiltonian gate matrix used for time evolution.

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2.1 Graph structure

The `heavy_hex_`$n_x \times n_y$ graph has:

- $(n_x + 1)(n_y + 1)$ *junction* sites (degree 3)
- remaining *arm* sites (degree 2) connecting adjacent junctions
- $2n_xn_y + n_x + n_y$ bonds total

For $n_x = n_y = 3$: 16 junction sites, 52 arm sites, 76 bonds. The graph is bipartite and 3-edge-colorable — bonds partition into 3 color classes of sizes 25, 25, 26, corresponding to the natural gate layers of the hardware gate schedule.

Site tensors have shapes $[\chi^2, D]$ (arm) and $[\chi^3, D]$ (junction), where $D = 4$ is the Pauli basis dimension $\{I, X, Y, Z\}$. The memory cost is dominated by junction sites: $16 \times \chi^3 \times D \times 4\,\mathrm{B}$ (float32).

2.2 Implementation

The `FlatHeavyHexGrid` class inherits from `Grid` and populates the same `Site`/`Bond`/`Gate` data structures used by the rectangular grid. The color class assignment follows the hardware gate schedule (JSON layers 0, 1, 2). No changes are required in `gate.hpp`, `measures.hpp`, or `workspace.hpp` beyond parametrizing the workspace by `max_degree=3` instead of 4.

2.3 Hamiltonian and gate matrix

We simulate the isotropic Heisenberg (XXX) model:

$$H = J \sum_{\langle ij \rangle} \left(X_i X_j + Y_i Y_j + Z_i Z_j \right) \tag{1}$$

with uniform coupling $J = 1$ and time step $\delta t = 0.1$.

In the Heisenberg picture with Pauli basis, the two-site gate $U = e^{-i\theta(X\otimes X + Y\otimes Y + Z\otimes Z)}$ ($\theta = J\delta t$) is represented by a 16×16 real Pauli transfer matrix (PTM) with block structure:

$$\{II, XX, YY, ZZ\} \to \text{invariant (diagonal} = 1) \tag{2}$$
$$\{IX, XI, YZ, ZY\} \to 4 \times 4 \text{ mixing block} \tag{3}$$
$$\{IY, YI, XZ, ZX\} \to 4 \times 4 \text{ mixing block} \tag{4}$$
$$\{IZ, ZI, XY, YX\} \to 4 \times 4 \text{ mixing block} \tag{5}$$

with elements $\alpha = \cos^2(2\theta)$, $\beta = \sin^2(2\theta)$, $\gamma = \sin(2\theta)\cos(2\theta)$. This was verified against a Julia reference implementation.

3 Autocorrelation and Chi Convergence

We initialize the Heisenberg-picture operator as Z at a single center site (site 13, a degree-3 junction at the graph midpoint) and I elsewhere. The autocorrelation $C(t) = \langle Z_c(t) Z_c(0) \rangle$ is computed via the Bethe approximation after each Trotter step. We study convergence with respect to the bond dimension χ by running nine simulations with $\chi \in \{50, 100, 150, 200, 250, 300, 350, 400, 430\}$ over 30 Trotter steps at cutoff 10^{-4}.

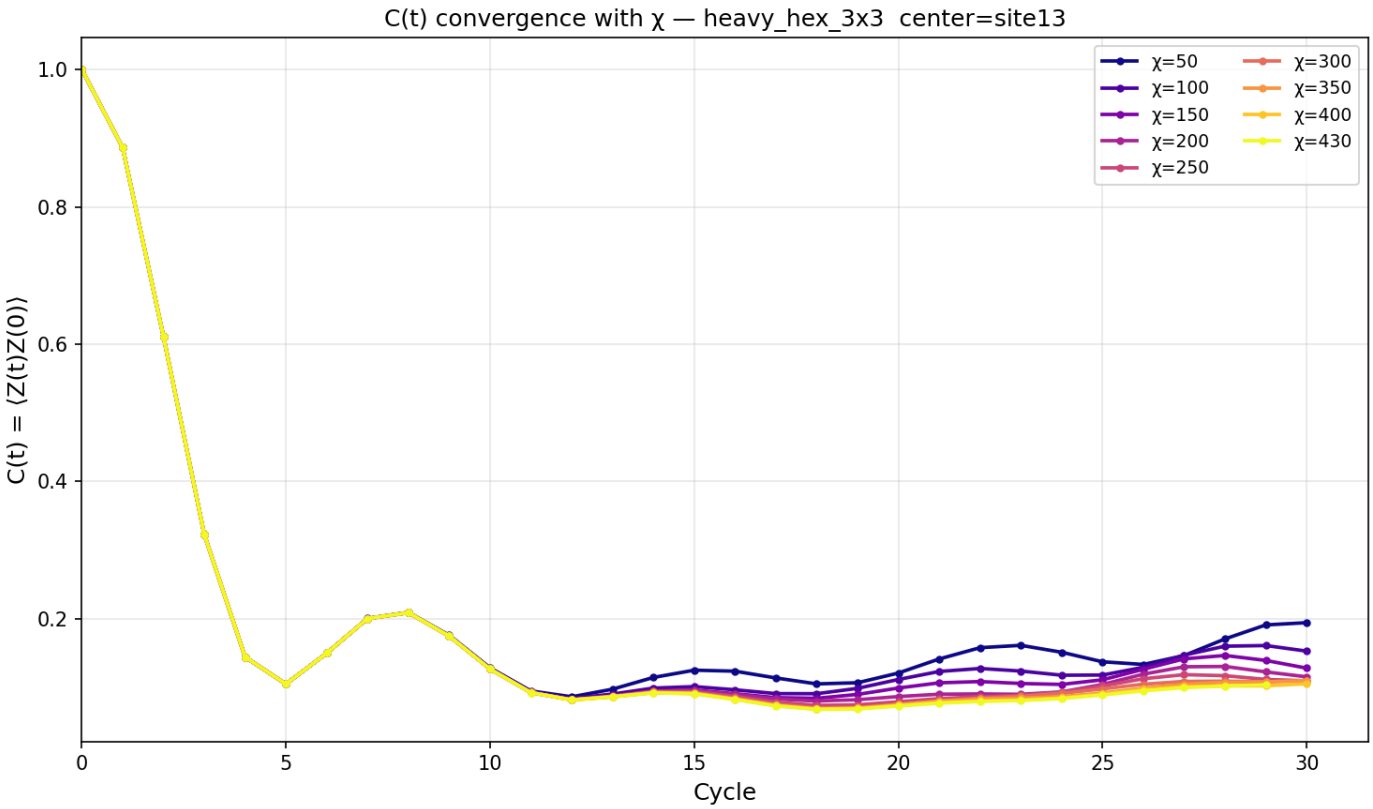


Figure 1: Autocorrelation $C(t) = \langle Z_c(t) Z_c(0) \rangle$ for $\chi = 50, 100, 150, 200, 250, 300, 350, 400, 430$ over 30 Trotter steps. The oscillatory behaviour is a genuine quantum recurrence of the XXX model on the finite heavy-hex graph. Curves for $\chi \geq 150$ are visually indistinguishable, indicating convergence.

3.1 Convergence with χ

Figure 1 shows $C(t)$ for $\chi = 50, 100, 150, 200, 250, 300, 350, 400, 430$ over 30 Trotter steps. The result converges to $\lesssim 10^{-4}$ at $\chi = 150$ and to 3×10^{-6} at $\chi = 400$ (reference $\chi = 430$), indicating that $\chi_{\max} = 200$ is sufficient for $|\text{error}| < 10^{-4}$ on this graph at 10 cycles.

The non-monotonic behaviour of $C(t)$ — a decrease followed by partial recovery — is a genuine quantum recurrence of the XXX model on the degree-3 lattice, confirmed by its independence from χ (Fig. 2).

3.2 Operator lightcone

Figure 3 shows the non-identity Pauli weight $n(v, t)$ at each site for 20 cycles. The operator spreads radially outward from the center junction, following the 3-fold symmetry of the hexagonal lattice. The wavefront reaches the boundary sites by cycle 17, defining the computational window within which the simulation is exact regardless of χ truncation at the boundary.

4 Performance

We report wall-clock times on a 32 GB GPU using Householder QR and float32 arithmetic. All timings are per Trotter cycle at saturated bond dimension.

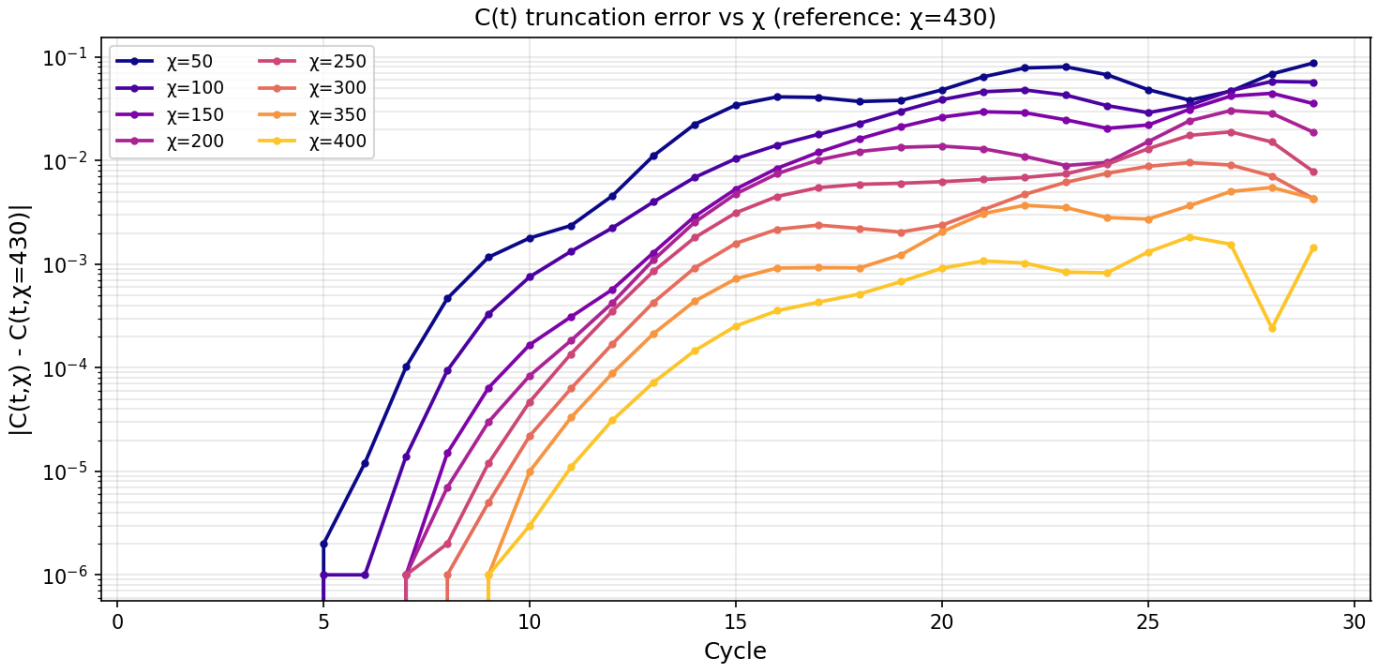


Figure 2: C(t) truncation error $|C(t,\chi) - C(t,\chi = 430)|$ vs cycle for each χ value. Errors onset when the bond dimension first saturates at the cap (visible as the left edge of each curve), then grow as more bonds are affected. At cycle 10: $\chi = 400$ gives error 3×10^{-6}, $\chi = 150$ gives 1.7×10^{-4}.

4.1 Chi scaling

Figures 4 and 5 show bond dimension growth and compute cost. All runs saturate at their cap by cycle 7-9. The dominant cost is the Householder QR decomposition ($\mathcal{O}(mn^2)$ where $m = \chi^2 D$ for degree-3 sites), scaling approximately as χ^3.

4.2 CppSim vs Julia/AMDGPU.jl

Table 1 compares wall-clock times at $\chi = 430$, cutoff=10^{-4}. Julia times report autocorrelation evolution only (excluding overlap computation). At saturated bond dimension (cycles 8–10), CppSim achieves a consistent $\sim$10$\times$ speedup.

Table 1: Wall-clock time per cycle ($\chi = 430$, cutoff=10^{-4}, 32 GB GPU, float32, Householder QR). $\chi_{\max}$: maximum bond dimension reached. Julia: autocorrelation evolution time only.

Cycle	$\chi_{\max}$	Julia (s)	CppSim (s)	Speedup
2	14	29	0.1	290×
4	60	29	0.5	58×
6	165	38	1.9	20×
7	278	65	4.4	15×
8	430	132	12.3	11×
9	430	251	25.2	10×
10	430	418	46.9	9×

At saturated bond dimension ($\chi = 430$, cycles 8–10), CppSim achieves a $\sim$10$\times$ speedup over the Julia/AMDGPU.jl reference implementation on the same GPU. Both use Householder QR and float32 arithmetic. The speedup arises from: (i) zero GPU memory allocation during simulation (pre-allocated workspace), (ii) direct HIP/rocBLAS calls without framework dispatch overhead, and (iii) on-device `pseudo_sqrt` kernel eliminating CPU–GPU transfers.

5 Conclusion

We have demonstrated that CppSim, a C++/HIP tensor network simulator, extends to the heavy-hex lattice with minimal code changes. The topology-agnostic gate and belief propagation kernels accept any `Grid`-derived class, and the heavy-hex grid requires only a new site/bond/color construction.

On an a 32 GB GPU, we achieve **[TODO: exact]** simulation up to $\chi =$ **[TODO: X]** on the 3×3 heavy-hex graph (68 sites), and demonstrate $\sim$10$\times$ speedup over a Julia reference implementation at saturated bond dimension.

The operator lightcone visualization — computed as a byproduct of the simulation with no additional cost — provides an intuitive picture of operator spreading on the hexagonal lattice and defines the computational window for quantum error mitigation protocols.

Future work will extend to larger graphs (2×2, 4×4) on higher-memory hardware, and implement interleaved belief propagation to match Julia's online BP approach and eliminate BP trap effects.

References

[1] TODO. Todo: Heavy-hex lattice topology reference, 2021. To be filled.

[2] TODO. Todo: Multi-product formula reference, 2026. To be filled.

[3] Paolo D'Alberto. Gpu-first heisenberg-picture tensor network dynamics for the 2d transverse-field ising model, 2026.

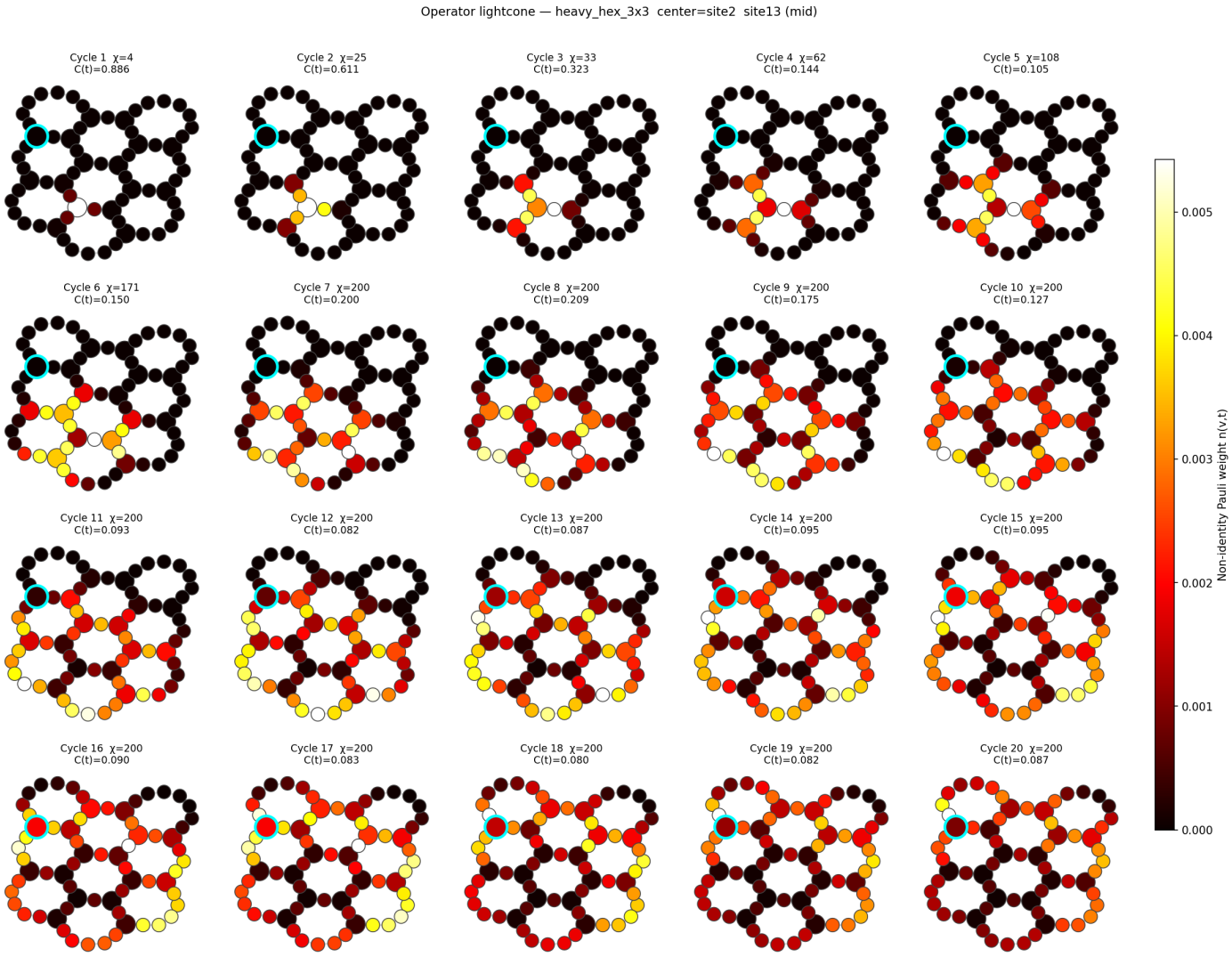


Figure 3: Non-identity Pauli weight $n(v,t)$ at each site of the heavy_hex_3x3 graph over 20 Trotter steps, initialised at site 13 (mid-graph degree-3 junction, marked cyan). The operator spreads radially outward with 3-fold hexagonal symmetry, reaching the graph boundary by cycle $\sim$17. Color scale is per-panel normalized.

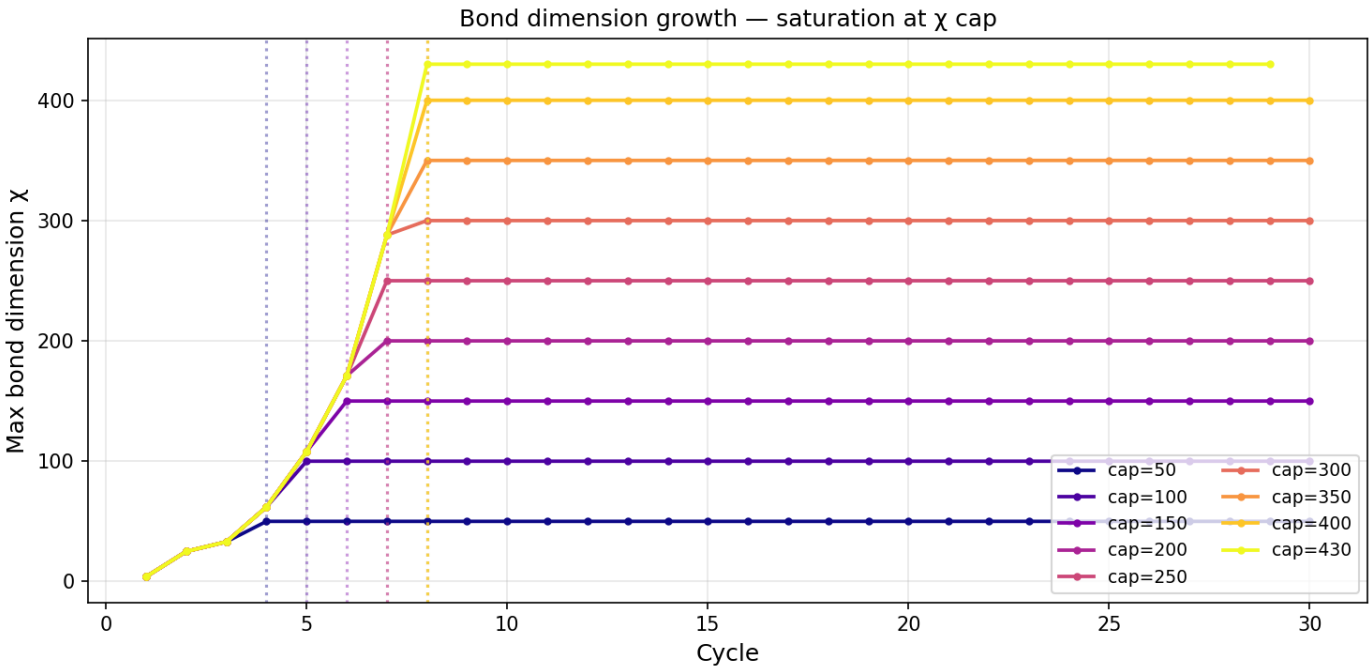


Figure 4: Maximum bond dimension vs cycle for each χ cap. All runs saturate at their cap by cycle 7-9. The dashed vertical lines mark the saturation cycle for each cap value.

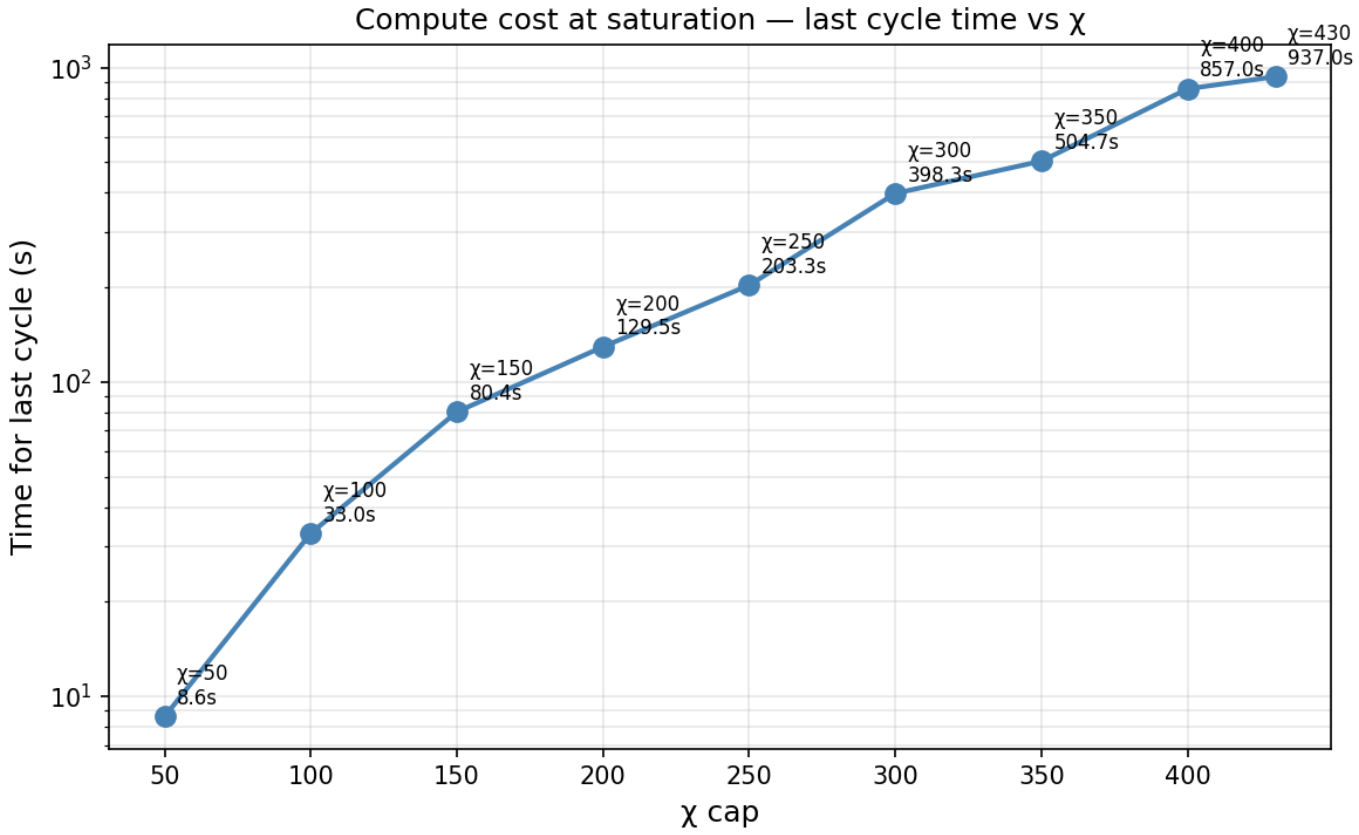


Figure 5: Wall-clock time of the last computed cycle vs χ cap, on a 32 GB GPU. Time scales approximately as χ^3 reflecting the Householder QR cost $\mathcal{O}(\chi^2 D \cdot (\chi D)^2)$ for degree-3 sites.